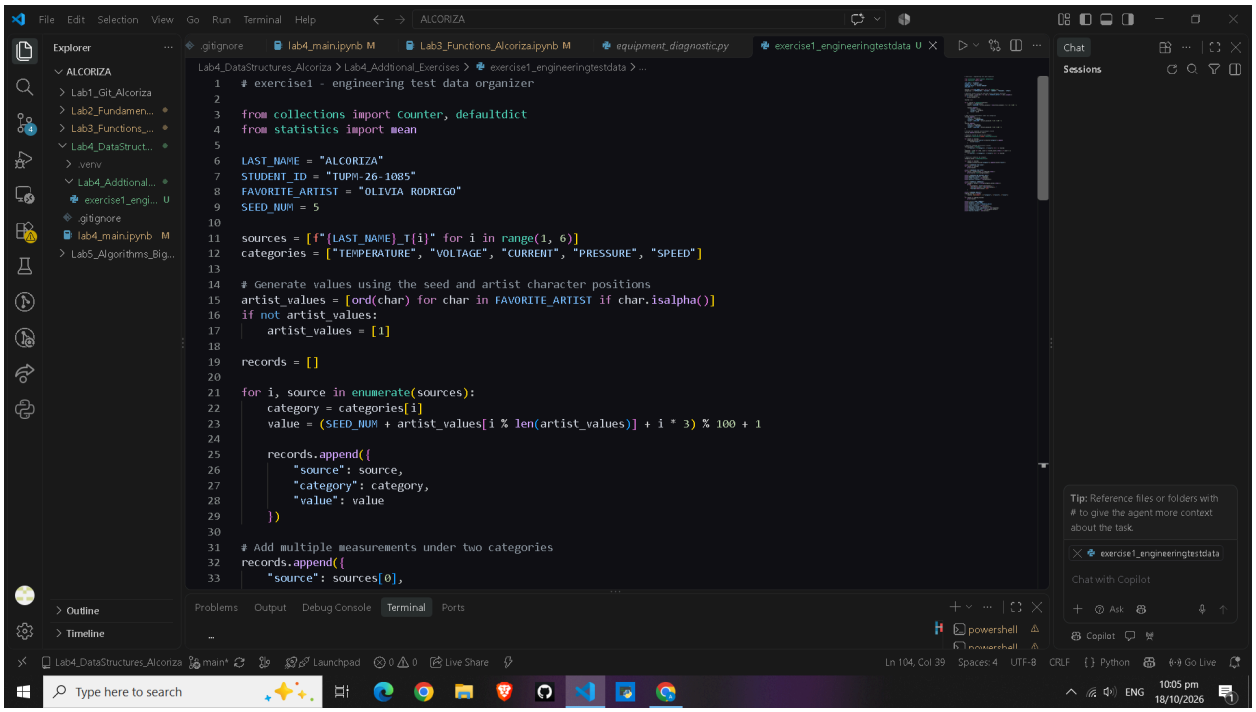
Avineth Alcoriza
BSECE -1C
COMPROG 1 - LAB 4 ADDITIONAL EXERCISES

EXERCISE 1 CODE



```python
# exercise1 - engineering test data organizer

from collections import Counter, defaultdict
from statistics import mean

LAST_NAME = "ALCORIZA"
STUDENT_ID = "TUPM-26-1085"
FAVORITE_ARTIST = "OLIVIA RODRIGO"
SEED_NUM = 5

sources = [f"{LAST_NAME}_T{i}" for i in range(1, 6)]
categories = ["TEMPERATURE", "VOLTAGE", "CURRENT", "PRESSURE", "SPEED"]

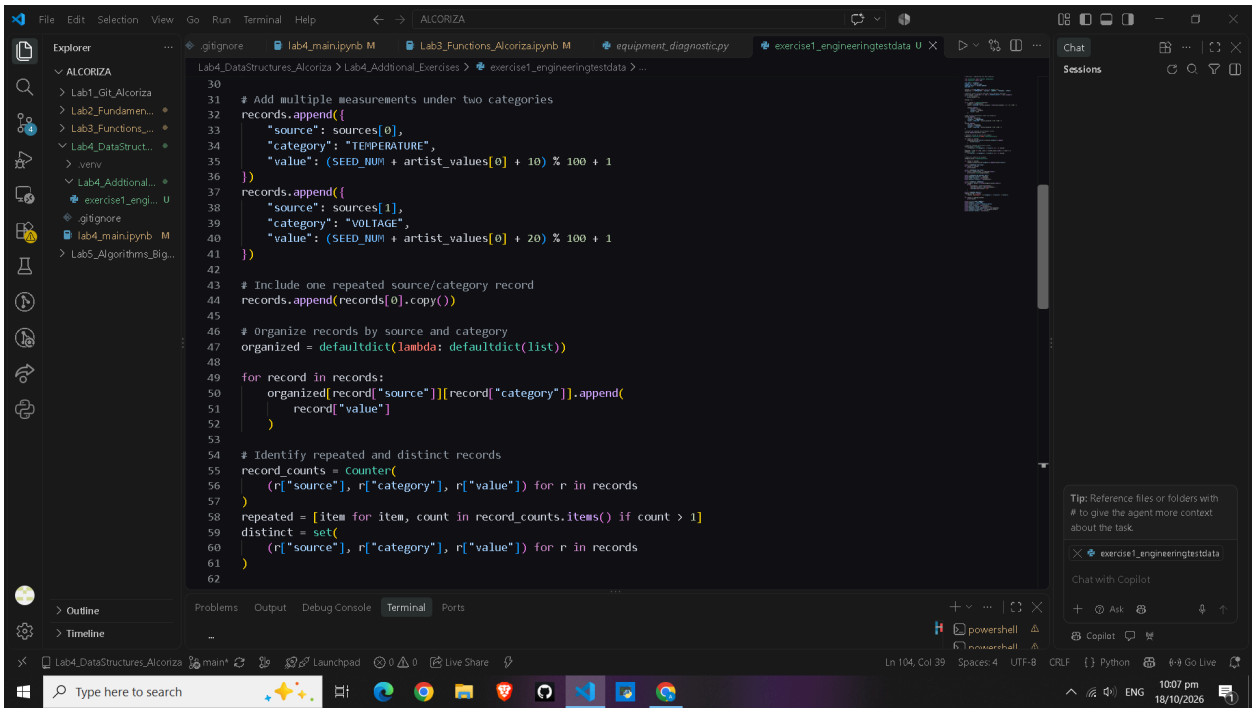
# Generate values using the seed and artist character positions
artist_values = [ord(char) for char in FAVORITE_ARTIST if char.isalpha()]
if not artist_values:
    artist_values = [1]

records = []

for i, source in enumerate(sources):
    category = categories[i]
    value = (SEED_NUM + artist_values[i % len(artist_values)] + i * 3) % 100 + 1

    records.append({
        "source": source,
        "category": category,
        "value": value
    })

# Add multiple measurements under two categories
records.append({
    "source": sources[0],
```



```python

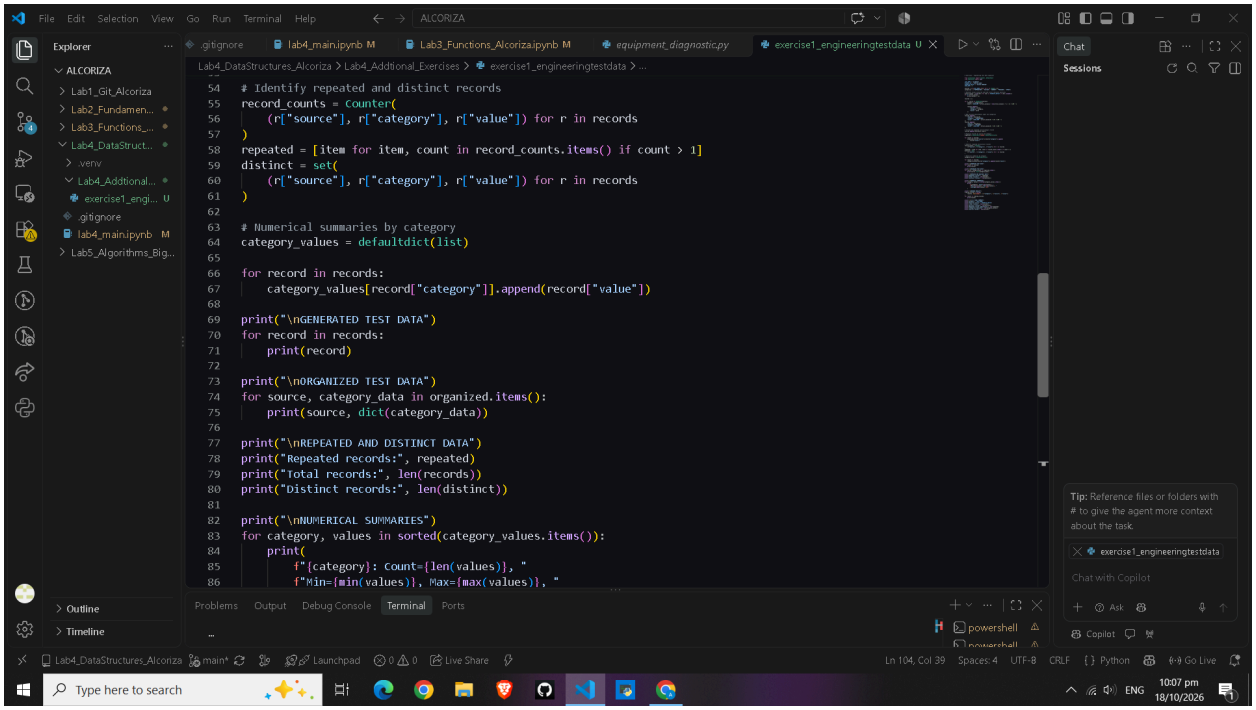
# Add multiple measurements under two categories
records.append({
    "source": sources[0],
    "category": "TEMPERATURE",
    "value": (SEED_NUM + artist_values[0] + 10) % 100 + 1
})
records.append({
    "source": sources[1],
    "category": "VOLTAGE",
    "value": (SEED_NUM + artist_values[0] + 20) % 100 + 1
})

# Include one repeated source/category record
records.append(records[0].copy())

# Organize records by source and category
organized = defaultdict(lambda: defaultdict(list))

for record in records:
    organized[record["source"]][record["category"]].append(
        record["value"]
    )

# Identify repeated and distinct records
record_counts = Counter(
    (r["source"], r["category"], r["value"]) for r in records
)
repeated = [item for item, count in record_counts.items() if count > 1]
distinct = set(
    (r["source"], r["category"], r["value"]) for r in records
)
```



```python
# Identify repeated and distinct records
record_counts = Counter(
    (r["source"], r["category"], r["value"]) for r in records
)
repeated = [item for item, count in record_counts.items() if count > 1]
distinct = set(
    (r["source"], r["category"], r["value"]) for r in records
)

# Numerical summaries by category
category_values = defaultdict(list)

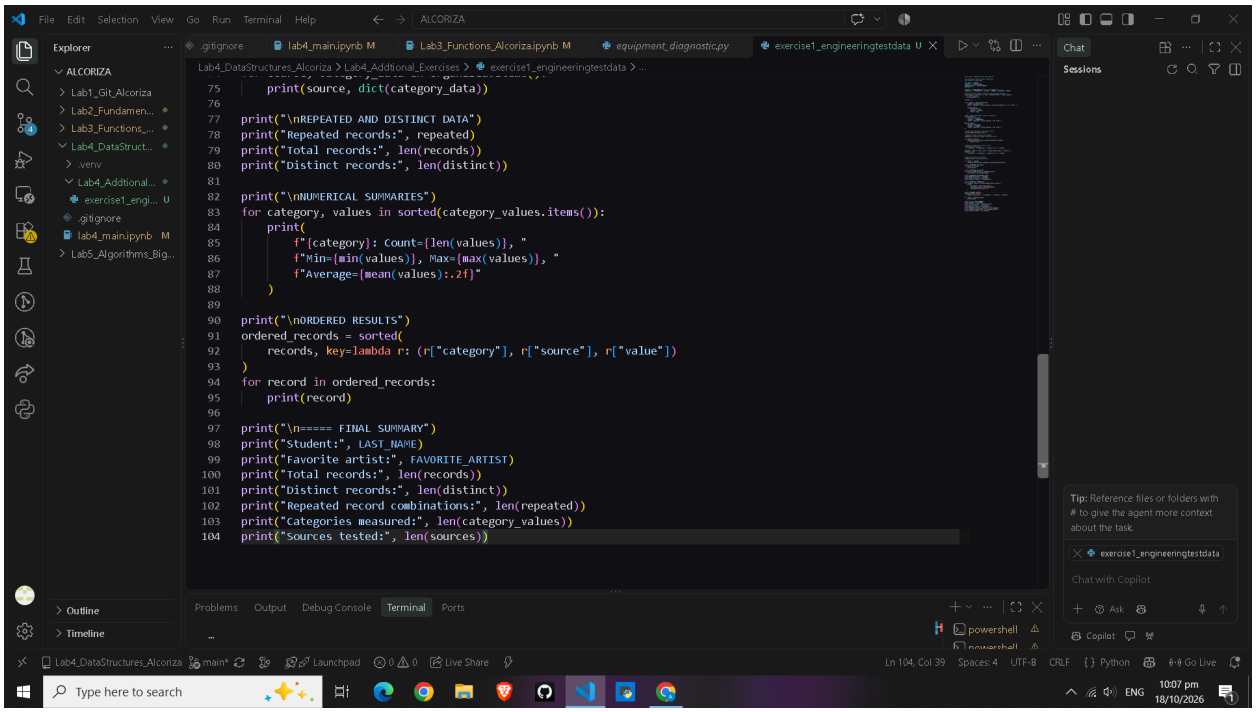
for record in records:
    category_values[record["category"]].append(record["value"])

print("\nGENERATED TEST DATA")
for record in records:
    print(record)

print("\nORGANIZED TEST DATA")
for source, category_data in organized.items():
    print(source, dict(category_data))

print("\nREPEATED AND DISTINCT DATA")
print("Repeated records:", repeated)
print("Total records:", len(records))
print("Distinct records:", len(distinct))

print("\nNUMERICAL SUMMARIES")
for category, values in sorted(category_values.items()):
    print(
        f"{category}: Count={len(values)}, "
        f"Min={min(values)}, Max={max(values)}, "
```



```python
    print(source, dict(category_data))

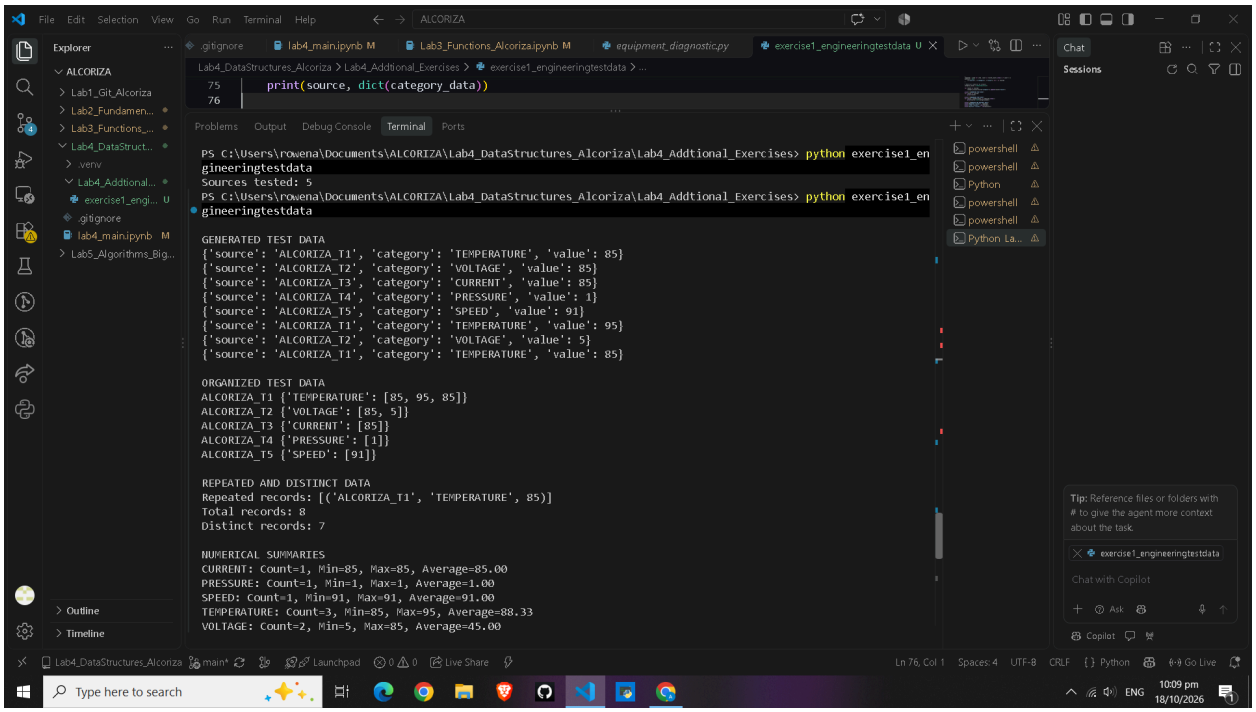
print("\nREPEATED AND DISTINCT DATA")
print("Repeated records:", repeated)
print("Total records:", len(records))
print("Distinct records:", len(distinct))

print("\nNUMERICAL SUMMARIES")
for category, values in sorted(category_values.items()):
    print(
        f"{category}: Count={len(values)}, "
        f"Min={min(values)}, Max={max(values)}, "
        f"Average={mean(values):.2f}"
    )

print("\nORDERED RESULTS")
ordered_records = sorted(
    records, key=lambda r: (r["category"], r["source"], r["value"])
)
for record in ordered_records:
    print(record)

print("\n===== FINAL SUMMARY")
print("Student:", LAST_NAME)
print("Favorite artist:", FAVORITE_ARTIST)
print("Total records:", len(records))
print("Distinct records:", len(distinct))
print("Repeated record combinations:", len(repeated))
print("Categories measured:", len(category_values))
print("Sources tested:", len(sources))
```

RESULTS



```
print(source, dict(category_data))

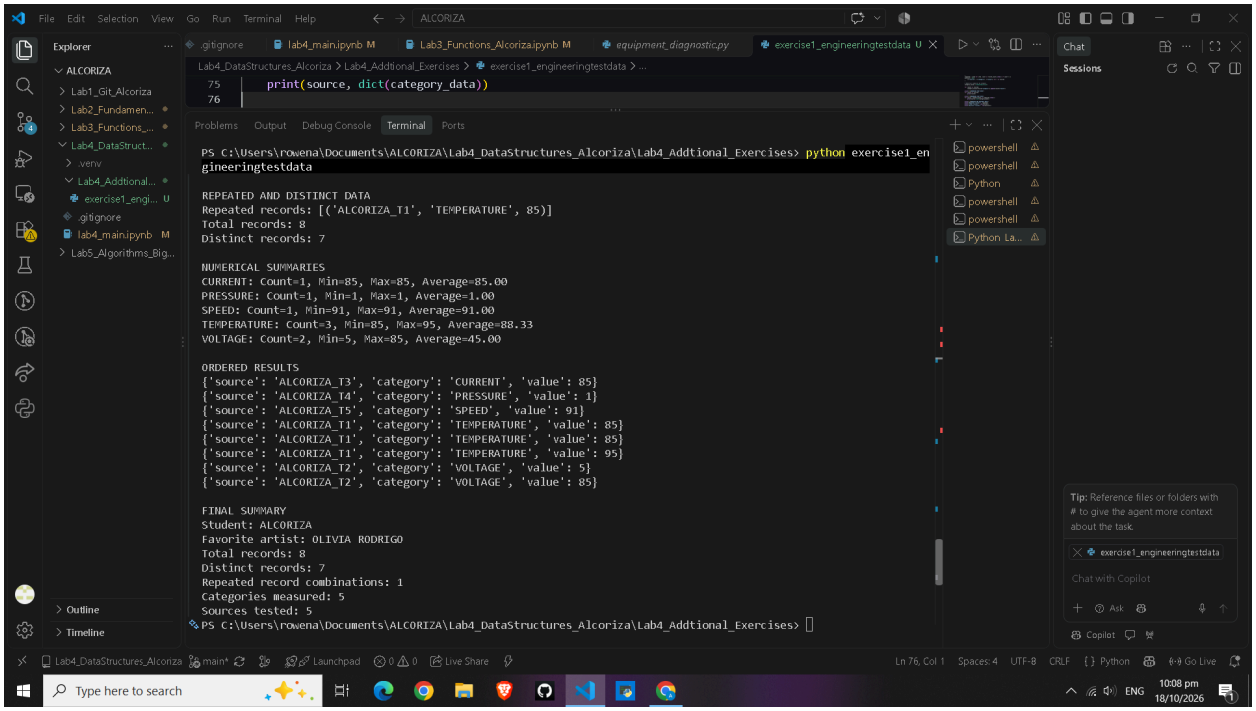
PS C:\Users\rowena\Documents\ALCORIZA\Lab4_DataStructures_Alcoriza\Lab4_Addtional_Exercises> python exercise1_engineeringtestdata
Sources tested: 5
PS C:\Users\rowena\Documents\ALCORIZA\Lab4_DataStructures_Alcoriza\Lab4_Addtional_Exercises> python exercise1_engineeringtestdata

GENERATED TEST DATA
{'source': 'ALCORIZA_T1', 'category': 'TEMPERATURE', 'value': 85}
{'source': 'ALCORIZA_T2', 'category': 'VOLTAGE', 'value': 85}
{'source': 'ALCORIZA_T3', 'category': 'CURRENT', 'value': 85}
{'source': 'ALCORIZA_T4', 'category': 'PRESSURE', 'value': 1}
{'source': 'ALCORIZA_T5', 'category': 'SPEED', 'value': 91}
{'source': 'ALCORIZA_T1', 'category': 'TEMPERATURE', 'value': 95}
{'source': 'ALCORIZA_T2', 'category': 'VOLTAGE', 'value': 5}
{'source': 'ALCORIZA_T1', 'category': 'TEMPERATURE', 'value': 85}

ORGANIZED TEST DATA
ALCORIZA_T1 {'TEMPERATURE': [85, 95, 85]}
ALCORIZA_T2 {'VOLTAGE': [85, 5]}
ALCORIZA_T3 {'CURRENT': [85]}
ALCORIZA_T4 {'PRESSURE': [1]}
ALCORIZA_T5 {'SPEED': [91]}

REPEATED AND DISTINCT DATA
Repeated records: [('ALCORIZA_T1', 'TEMPERATURE', 85)]
Total records: 8
Distinct records: 7

NUMERICAL SUMMARIES
CURRENT: Count=1, Min=85, Max=85, Average=85.00
PRESSURE: Count=1, Min=1, Max=1, Average=1.00
SPEED: Count=1, Min=91, Max=91, Average=91.00
TEMPERATURE: Count=3, Min=85, Max=95, Average=88.33
VOLTAGE: Count=2, Min=5, Max=85, Average=45.00
```



```
print(source, dict(category_data))

PS C:\Users\rowena\Documents\ALCORIZA\Lab4_DataStructures_Alcoriza\Lab4_Addtional_Exercises> python exercise1_engineeringtestdata

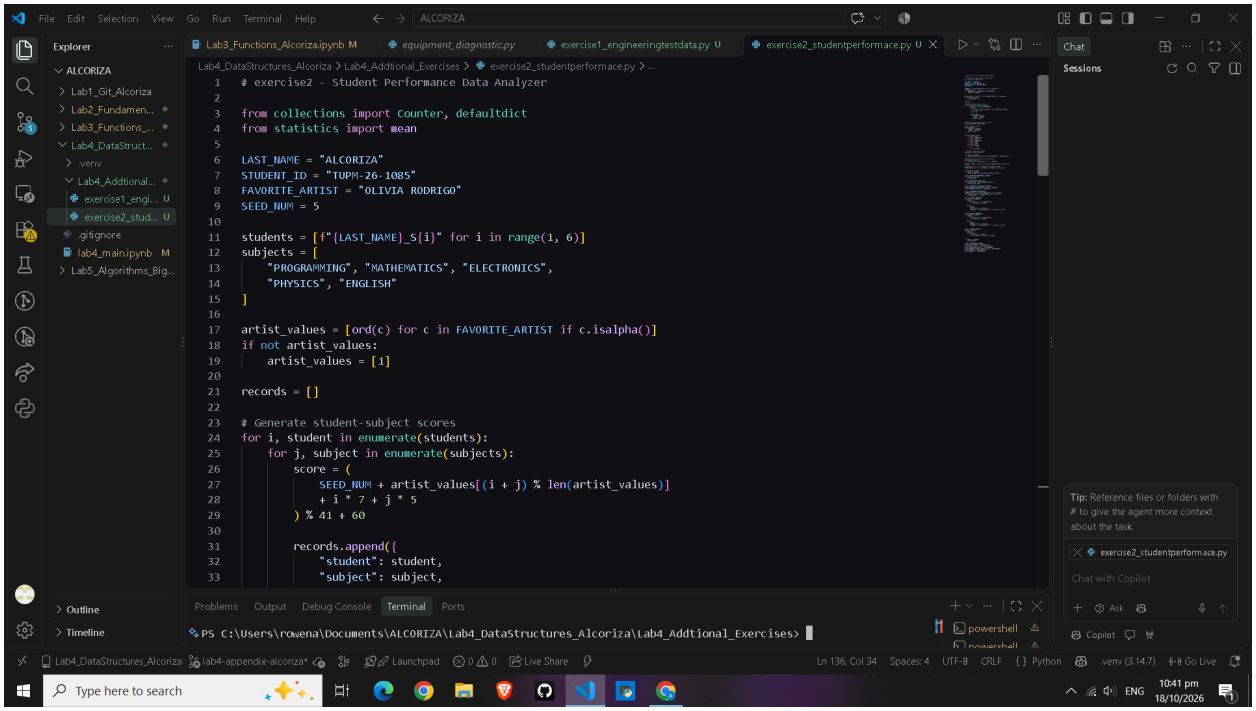
REPEATED AND DISTINCT DATA
Repeated records: [('ALCORIZA_T1', 'TEMPERATURE', 85)]
Total records: 8
Distinct records: 7

NUMERICAL SUMMARIES
CURRENT: Count=1, Min=85, Max=85, Average=85.00
PRESSURE: Count=1, Min=1, Max=1, Average=1.00
SPEED: Count=1, Min=91, Max=91, Average=91.00
TEMPERATURE: Count=3, Min=85, Max=95, Average=88.33
VOLTAGE: Count=2, Min=5, Max=85, Average=45.00

ORDERED RESULTS
{'source': 'ALCORIZA_T3', 'category': 'CURRENT', 'value': 85}
{'source': 'ALCORIZA_T4', 'category': 'PRESSURE', 'value': 1}
{'source': 'ALCORIZA_T5', 'category': 'SPEED', 'value': 91}
{'source': 'ALCORIZA_T1', 'category': 'TEMPERATURE', 'value': 85}
{'source': 'ALCORIZA_T1', 'category': 'TEMPERATURE', 'value': 85}
{'source': 'ALCORIZA_T1', 'category': 'TEMPERATURE', 'value': 95}
{'source': 'ALCORIZA_T2', 'category': 'VOLTAGE', 'value': 5}
{'source': 'ALCORIZA_T2', 'category': 'VOLTAGE', 'value': 85}

FINAL SUMMARY
Student: ALCORIZA
Favorite artist: OLIVIA RODRIGO
Total records: 8
Distinct records: 7
Repeated record combinations: 1
Categories measured: 5
Sources tested: 5
PS C:\Users\rowena\Documents\ALCORIZA\Lab4_DataStructures_Alcoriza\Lab4_Addtional_Exercises>
```

EXERCISE 2
CODE



```python
# exercise2 - Student Performance Data Analyzer

from collections import Counter, defaultdict
from statistics import mean

LAST_NAME = "ALCORIZA"
STUDENT_ID = "TUPM-26-1085"
FAVORITE_ARTIST = "OLIVIA RODRIGO"
SEED_NUM = 5

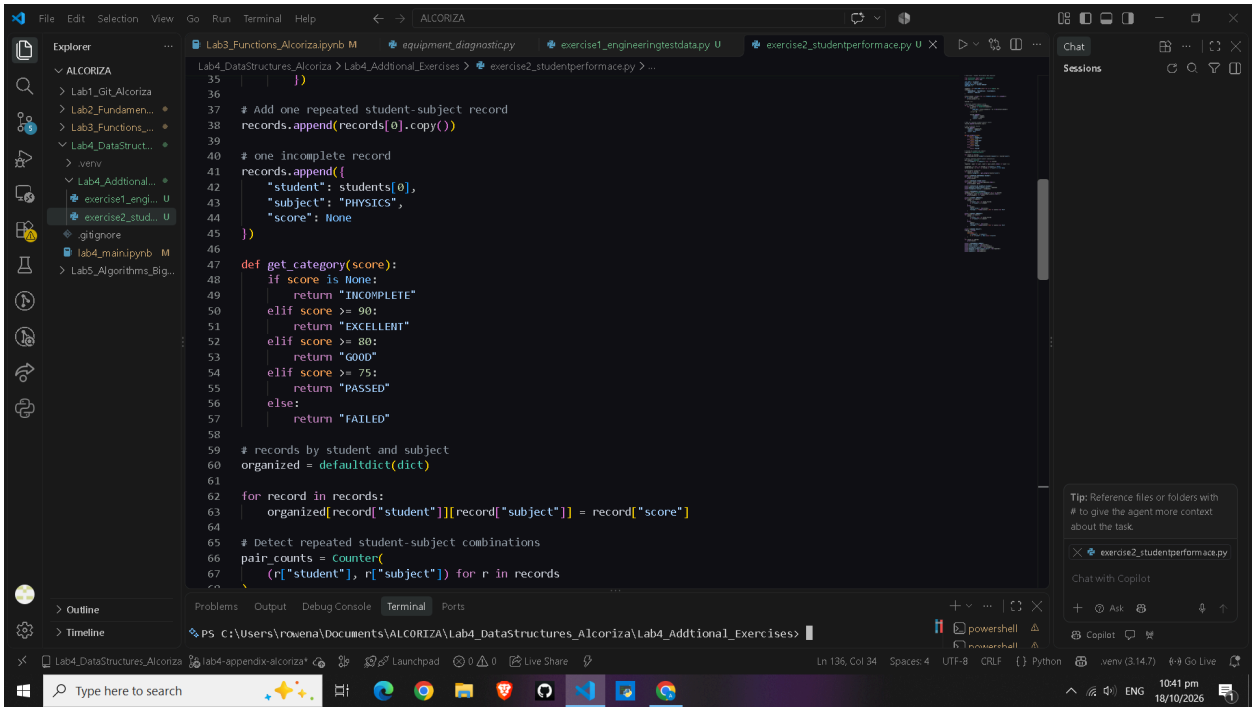
students = [f"{LAST_NAME}_S{i}" for i in range(1, 6)]
subjects = [
    "PROGRAMMING", "MATHEMATICS", "ELECTRONICS",
    "PHYSICS", "ENGLISH"
]

artist_values = [ord(c) for c in FAVORITE_ARTIST if c.isalpha()]
if not artist_values:
    artist_values = [1]

records = []

# Generate student-subject scores
for i, student in enumerate(students):
    for j, subject in enumerate(subjects):
        score = (
            SEED_NUM + artist_values[(i + j) % len(artist_values)]
            + i * 7 + j * 5
        ) % 41 + 60

        records.append({
            "student": student,
            "subject": subject,
```



```python
        })

# Add one repeated student-subject record
records.append(records[0].copy())

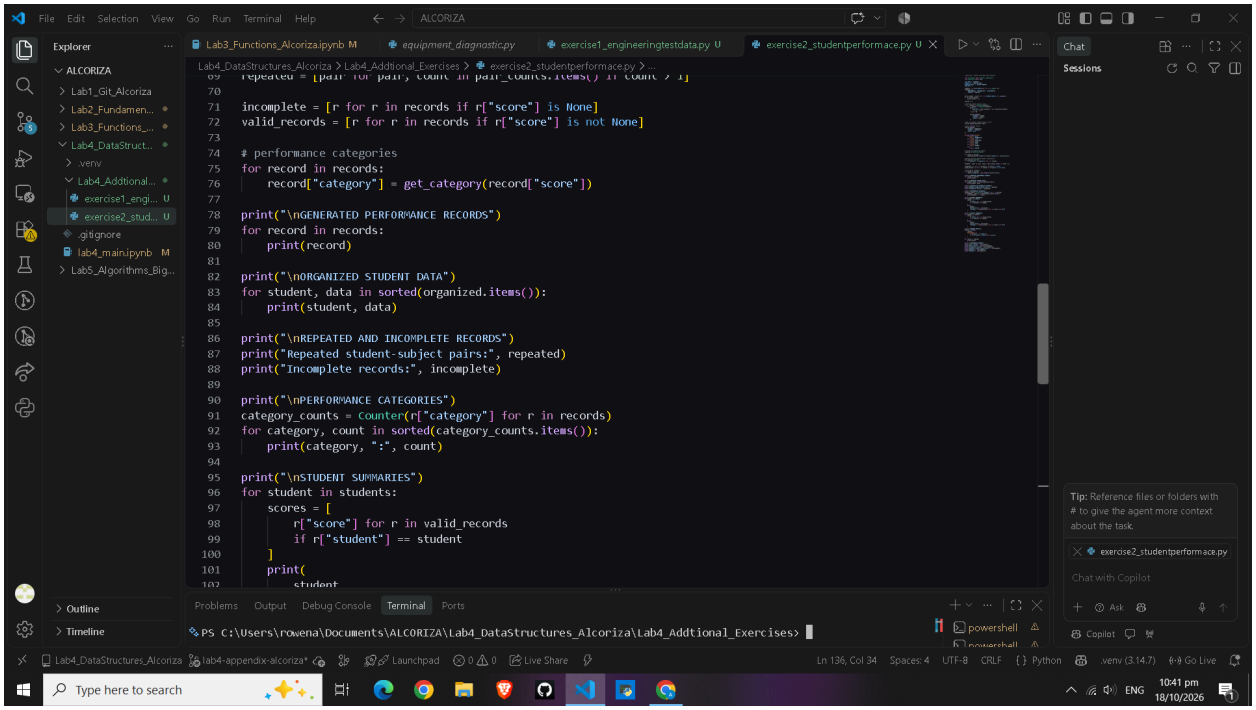
# one incomplete record
records.append({
    "student": students[0],
    "subject": "PHYSICS",
    "score": None
})

def get_category(score):
    if score is None:
        return "INCOMPLETE"
    elif score >= 90:
        return "EXCELLENT"
    elif score >= 80:
        return "GOOD"
    elif score >= 75:
        return "PASSED"
    else:
        return "FAILED"

# records by student and subject
organized = defaultdict(dict)

for record in records:
    organized[record["student"]][record["subject"]] = record["score"]

# Detect repeated student-subject combinations
pair_counts = Counter(
    (r["student"], r["subject"]) for r in records
```



```
incomplete = [r for r in records if r["score"] is None]
valid_records = [r for r in records if r["score"] is not None]

# performance categories
for record in records:
    record["category"] = get_category(record["score"])

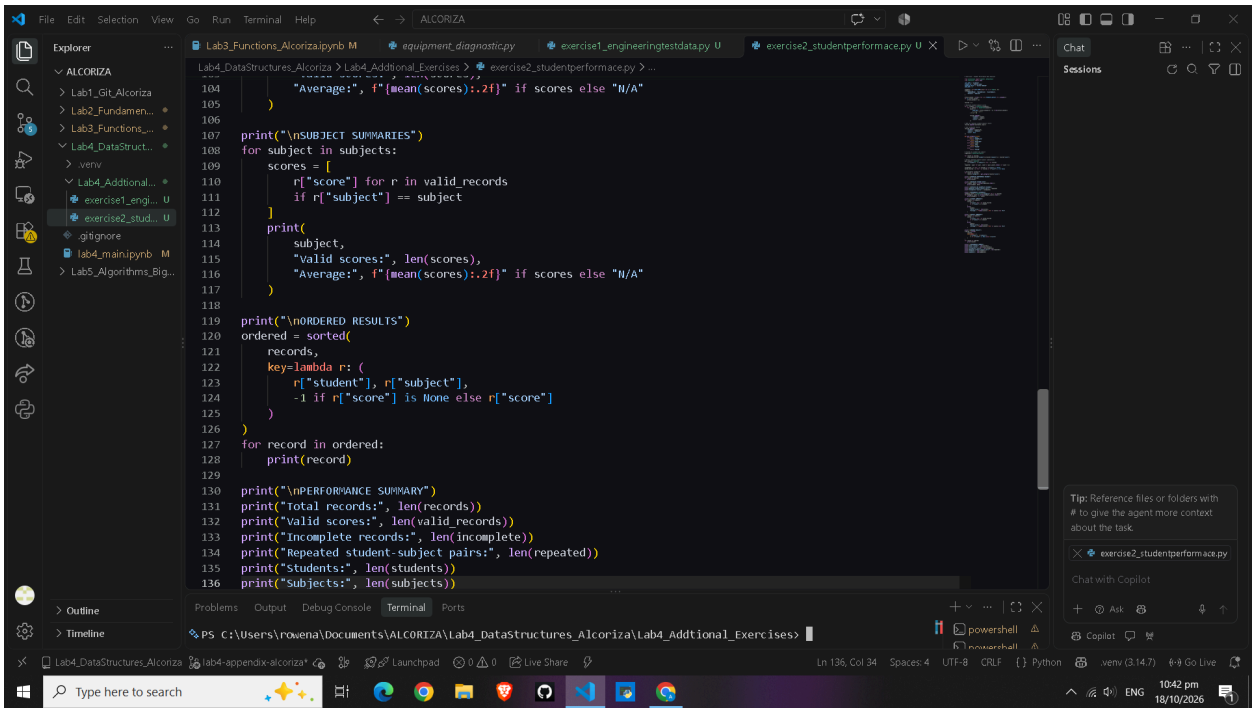
print("\nGENERATED PERFORMANCE RECORDS")
for record in records:
    print(record)

print("\nORGANIZED STUDENT DATA")
for student, data in sorted(organized.items()):
    print(student, data)

print("\nREPEATED AND INCOMPLETE RECORDS")
print("Repeated student-subject pairs:", repeated)
print("Incomplete records:", incomplete)

print("\nPERFORMANCE CATEGORIES")
category_counts = Counter(r["category"] for r in records)
for category, count in sorted(category_counts.items()):
    print(category, ":", count)

print("\nSTUDENT SUMMARIES")
for student in students:
    scores = [
        r["score"] for r in valid_records
        if r["student"] == student
    ]
    print(
        student
```



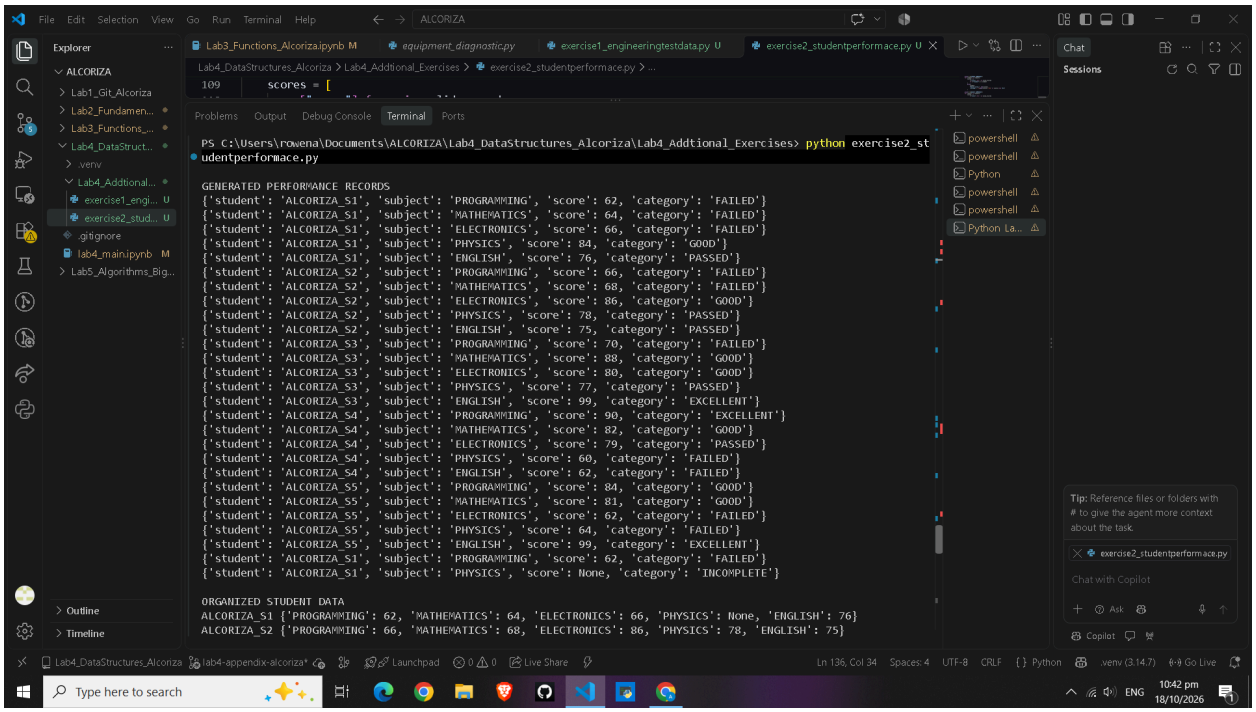
```
        "Average:", f"{mean(scores):.2f}" if scores else "N/A"
    )

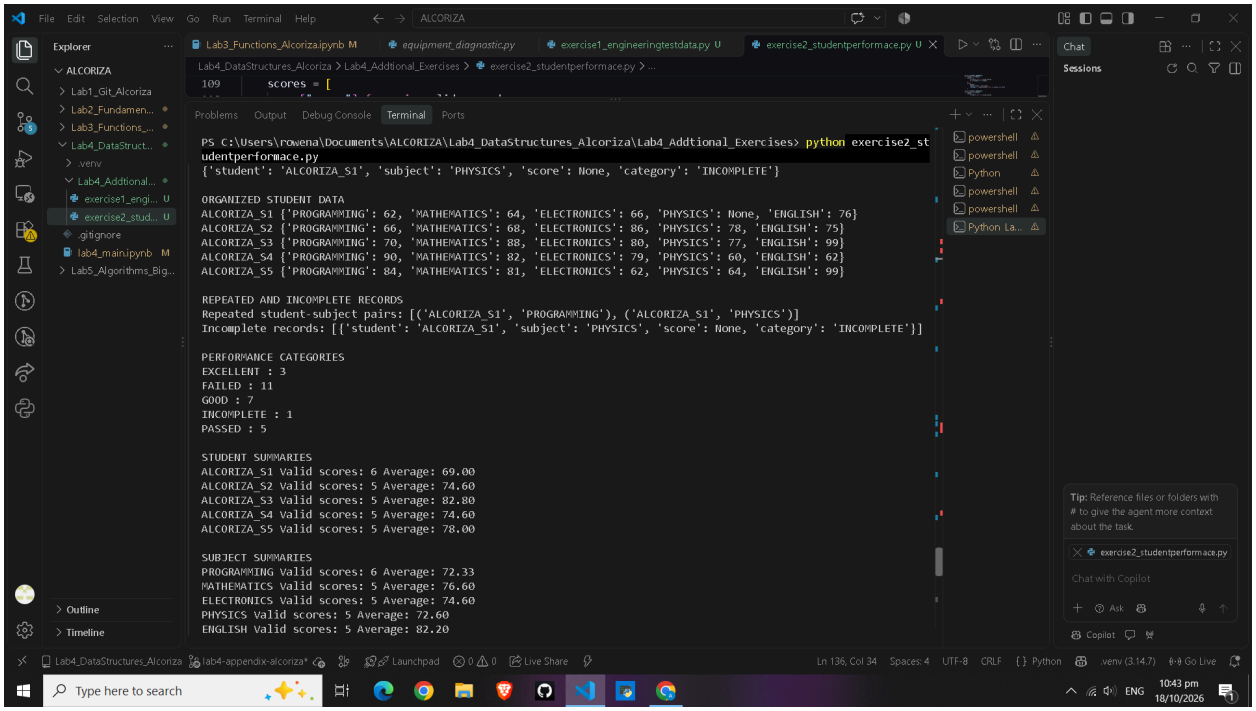
print("\nSUBJECT SUMMARIES")
for subject in subjects:
    scores = [
        r["score"] for r in valid_records
        if r["subject"] == subject
    ]
    print(
        subject,
        "Valid scores:", len(scores),
        "Average:", f"{mean(scores):.2f}" if scores else "N/A"
    )

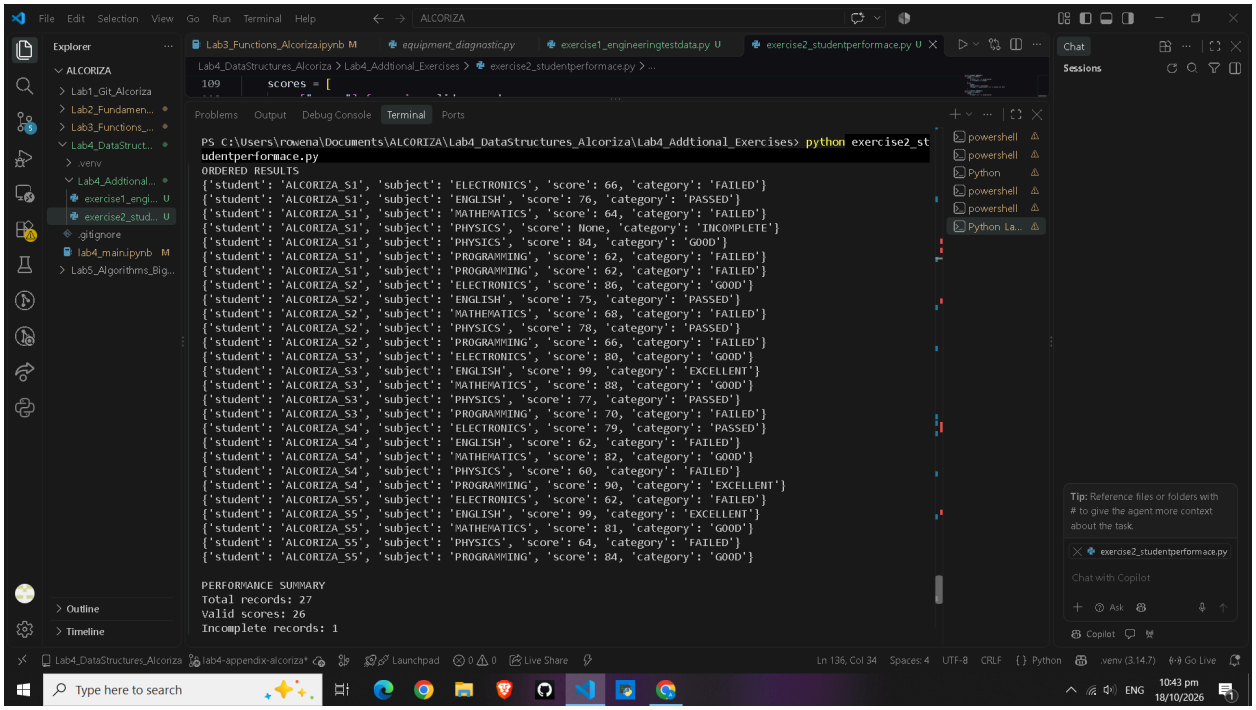
print("\nORDERED RESULTS")
ordered = sorted(
    records,
    key=lambda r: (
        r["student"], r["subject"],
        -1 if r["score"] is None else r["score"]
    )
)
for record in ordered:
    print(record)

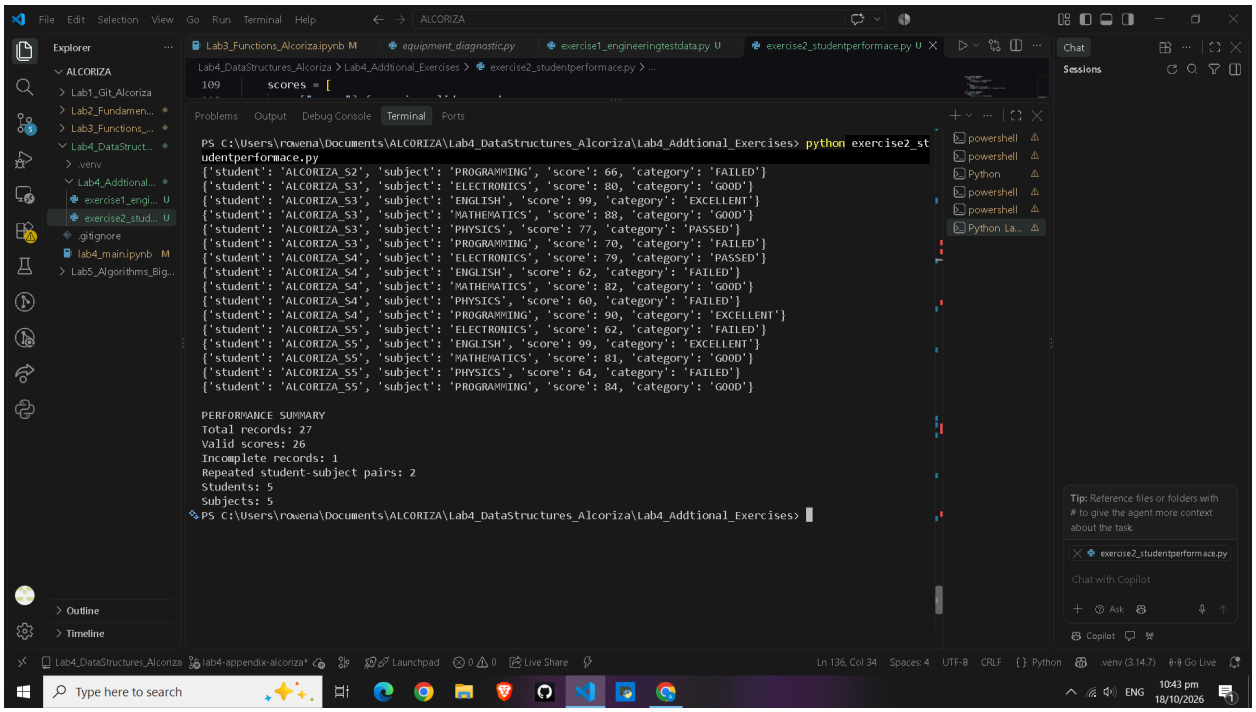
print("\nPERFORMANCE SUMMARY")
print("Total records:", len(records))
print("Valid scores:", len(valid_records))
print("Incomplete records:", len(incomplete))
print("Repeated student-subject pairs:", len(repeated))
print("Students:", len(students))
print("Subjects:", len(subjects))
```

RESULTS

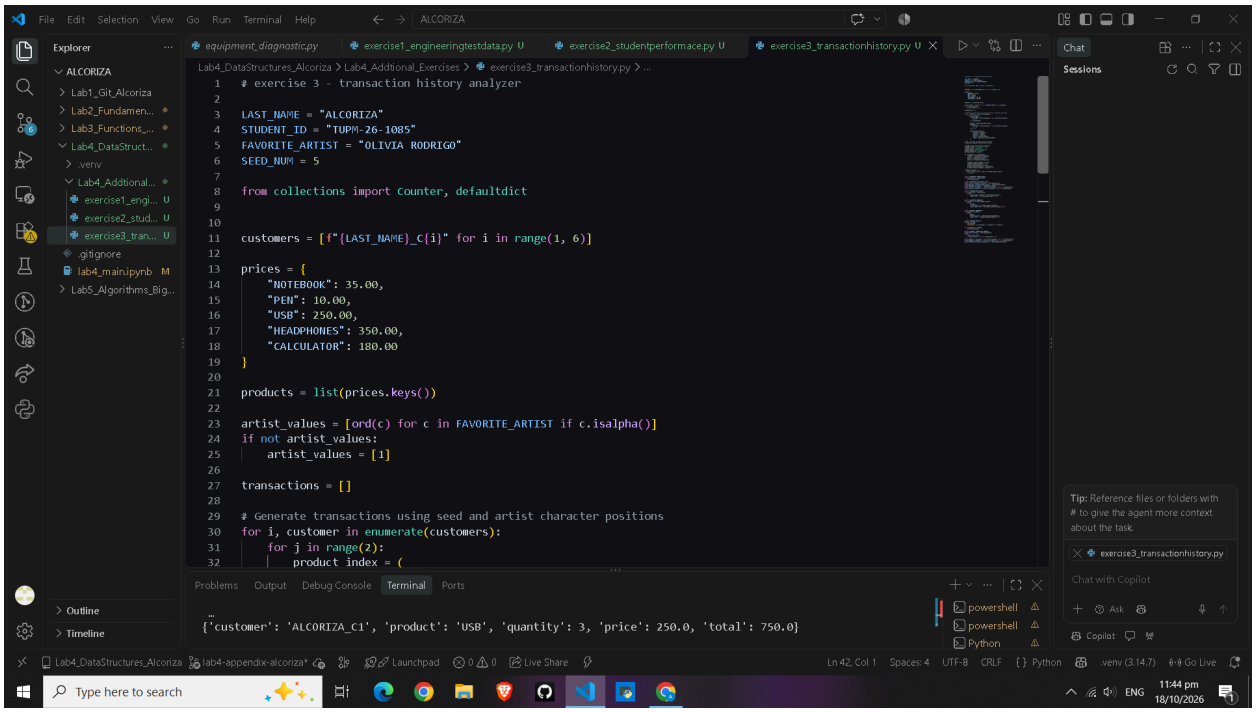

PS C:\Users\rowena\Documents\ALCORIZA\Lab4_DataStructures_Alcoriza\Lab4_Addtional_Exercises> python exercise2_studentperformace.py
GENERATED PERFORMANCE RECORDS
{'student': 'ALCORIZA_S1', 'subject': 'PROGRAMMING', 'score': 62, 'category': 'FAILED'}
{'student': 'ALCORIZA_S1', 'subject': 'MATHEMATICS', 'score': 64, 'category': 'FAILED'}
{'student': 'ALCORIZA_S1', 'subject': 'ELECTRONICS', 'score': 66, 'category': 'FAILED'}
{'student': 'ALCORIZA_S1', 'subject': 'PHYSICS', 'score': 84, 'category': 'GOOD'}
{'student': 'ALCORIZA_S1', 'subject': 'ENGLISH', 'score': 76, 'category': 'PASSED'}
{'student': 'ALCORIZA_S2', 'subject': 'PROGRAMMING', 'score': 66, 'category': 'FAILED'}
{'student': 'ALCORIZA_S2', 'subject': 'MATHEMATICS', 'score': 68, 'category': 'FAILED'}
{'student': 'ALCORIZA_S2', 'subject': 'ELECTRONICS', 'score': 86, 'category': 'GOOD'}
{'student': 'ALCORIZA_S2', 'subject': 'PHYSICS', 'score': 78, 'category': 'PASSED'}
{'student': 'ALCORIZA_S2', 'subject': 'ENGLISH', 'score': 75, 'category': 'PASSED'}
{'student': 'ALCORIZA_S3', 'subject': 'PROGRAMMING', 'score': 70, 'category': 'FAILED'}
{'student': 'ALCORIZA_S3', 'subject': 'MATHEMATICS', 'score': 88, 'category': 'GOOD'}
{'student': 'ALCORIZA_S3', 'subject': 'ELECTRONICS', 'score': 80, 'category': 'GOOD'}
{'student': 'ALCORIZA_S3', 'subject': 'PHYSICS', 'score': 77, 'category': 'PASSED'}
{'student': 'ALCORIZA_S3', 'subject': 'ENGLISH', 'score': 99, 'category': 'EXCELLENT'}
{'student': 'ALCORIZA_S4', 'subject': 'PROGRAMMING', 'score': 90, 'category': 'EXCELLENT'}
{'student': 'ALCORIZA_S4', 'subject': 'MATHEMATICS', 'score': 82, 'category': 'GOOD'}
{'student': 'ALCORIZA_S4', 'subject': 'ELECTRONICS', 'score': 79, 'category': 'PASSED'}
{'student': 'ALCORIZA_S4', 'subject': 'PHYSICS', 'score': 60, 'category': 'FAILED'}
{'student': 'ALCORIZA_S4', 'subject': 'ENGLISH', 'score': 62, 'category': 'FAILED'}
{'student': 'ALCORIZA_S5', 'subject': 'PROGRAMMING', 'score': 84, 'category': 'GOOD'}
{'student': 'ALCORIZA_S5', 'subject': 'MATHEMATICS', 'score': 81, 'category': 'GOOD'}
{'student': 'ALCORIZA_S5', 'subject': 'ELECTRONICS', 'score': 62, 'category': 'FAILED'}
{'student': 'ALCORIZA_S5', 'subject': 'PHYSICS', 'score': 64, 'category': 'FAILED'}
{'student': 'ALCORIZA_S5', 'subject': 'ENGLISH', 'score': 99, 'category': 'EXCELLENT'}
{'student': 'ALCORIZA_S1', 'subject': 'PROGRAMMING', 'score': 62, 'category': 'FAILED'}
{'student': 'ALCORIZA_S1', 'subject': 'PHYSICS', 'score': None, 'category': 'INCOMPLETE'}
ORGANIZED STUDENT DATA
ALCORIZA_S1 {'PROGRAMMING': 62, 'MATHEMATICS': 64, 'ELECTRONICS': 66, 'PHYSICS': None, 'ENGLISH': 76}
ALCORIZA_S2 {'PROGRAMMING': 66, 'MATHEMATICS': 68, 'ELECTRONICS': 86, 'PHYSICS': 78, 'ENGLISH': 75}


PS C:\Users\rowena\Documents\ALCORIZA\Lab4_DataStructures_Alcoriza\Lab4_Addtional_Exercises> python exercise2_studentperformace.py
{'student': 'ALCORIZA_S1', 'subject': 'PHYSICS', 'score': None, 'category': 'INCOMPLETE'}
ORGANIZED STUDENT DATA
ALCORIZA_S1 {'PROGRAMMING': 62, 'MATHEMATICS': 64, 'ELECTRONICS': 66, 'PHYSICS': None, 'ENGLISH': 76}
ALCORIZA_S2 {'PROGRAMMING': 66, 'MATHEMATICS': 68, 'ELECTRONICS': 86, 'PHYSICS': 78, 'ENGLISH': 75}
ALCORIZA_S3 {'PROGRAMMING': 70, 'MATHEMATICS': 88, 'ELECTRONICS': 80, 'PHYSICS': 77, 'ENGLISH': 99}
ALCORIZA_S4 {'PROGRAMMING': 90, 'MATHEMATICS': 82, 'ELECTRONICS': 79, 'PHYSICS': 60, 'ENGLISH': 62}
ALCORIZA_S5 {'PROGRAMMING': 84, 'MATHEMATICS': 81, 'ELECTRONICS': 62, 'PHYSICS': 64, 'ENGLISH': 99}
REPEATED AND INCOMPLETE RECORDS
Repeated student-subject pairs: [('ALCORIZA_S1', 'PROGRAMMING'), ('ALCORIZA_S1', 'PHYSICS')]
Incomplete records: [{'student': 'ALCORIZA_S1', 'subject': 'PHYSICS', 'score': None, 'category': 'INCOMPLETE'}]
PERFORMANCE CATEGORIES
EXCELLENT : 3
FAILED : 11
GOOD : 7
INCOMPLETE : 1
PASSED : 5
STUDENT SUMMARIES
ALCORIZA_S1 Valid scores: 6 Average: 69.00
ALCORIZA_S2 Valid scores: 5 Average: 74.60
ALCORIZA_S3 Valid scores: 5 Average: 82.80
ALCORIZA_S4 Valid scores: 5 Average: 74.60
ALCORIZA_S5 Valid scores: 5 Average: 78.00
SUBJECT SUMMARIES
PROGRAMMING Valid scores: 6 Average: 72.33
MATHEMATICS Valid scores: 5 Average: 76.60
ELECTRONICS Valid scores: 5 Average: 74.60
PHYSICS Valid scores: 5 Average: 72.60
ENGLISH Valid scores: 5 Average: 82.20


PS C:\Users\rowena\Documents\ALCORIZA\Lab4_DataStructures_Alcoriza\Lab4_Addtional_Exercises> python exercise2_studentperformace.py
ORDERED RESULTS
{'student': 'ALCORIZA_S1', 'subject': 'ELECTRONICS', 'score': 66, 'category': 'FAILED'}
{'student': 'ALCORIZA_S1', 'subject': 'ENGLISH', 'score': 76, 'category': 'PASSED'}
{'student': 'ALCORIZA_S1', 'subject': 'MATHEMATICS', 'score': 64, 'category': 'FAILED'}
{'student': 'ALCORIZA_S1', 'subject': 'PHYSICS', 'score': None, 'category': 'INCOMPLETE'}
{'student': 'ALCORIZA_S1', 'subject': 'PHYSICS', 'score': 84, 'category': 'GOOD'}
{'student': 'ALCORIZA_S1', 'subject': 'PROGRAMMING', 'score': 62, 'category': 'FAILED'}
{'student': 'ALCORIZA_S1', 'subject': 'PROGRAMMING', 'score': 62, 'category': 'FAILED'}
{'student': 'ALCORIZA_S2', 'subject': 'ELECTRONICS', 'score': 86, 'category': 'GOOD'}
{'student': 'ALCORIZA_S2', 'subject': 'ENGLISH', 'score': 75, 'category': 'PASSED'}
{'student': 'ALCORIZA_S2', 'subject': 'MATHEMATICS', 'score': 68, 'category': 'FAILED'}
{'student': 'ALCORIZA_S2', 'subject': 'PHYSICS', 'score': 78, 'category': 'PASSED'}
{'student': 'ALCORIZA_S2', 'subject': 'PROGRAMMING', 'score': 66, 'category': 'FAILED'}
{'student': 'ALCORIZA_S3', 'subject': 'ELECTRONICS', 'score': 80, 'category': 'GOOD'}
{'student': 'ALCORIZA_S3', 'subject': 'ENGLISH', 'score': 99, 'category': 'EXCELLENT'}
{'student': 'ALCORIZA_S3', 'subject': 'MATHEMATICS', 'score': 88, 'category': 'GOOD'}
{'student': 'ALCORIZA_S3', 'subject': 'PHYSICS', 'score': 77, 'category': 'PASSED'}
{'student': 'ALCORIZA_S3', 'subject': 'PROGRAMMING', 'score': 70, 'category': 'FAILED'}
{'student': 'ALCORIZA_S4', 'subject': 'ELECTRONICS', 'score': 79, 'category': 'PASSED'}
{'student': 'ALCORIZA_S4', 'subject': 'ENGLISH', 'score': 62, 'category': 'FAILED'}
{'student': 'ALCORIZA_S4', 'subject': 'MATHEMATICS', 'score': 82, 'category': 'GOOD'}
{'student': 'ALCORIZA_S4', 'subject': 'PHYSICS', 'score': 60, 'category': 'FAILED'}
{'student': 'ALCORIZA_S4', 'subject': 'PROGRAMMING', 'score': 90, 'category': 'EXCELLENT'}
{'student': 'ALCORIZA_S5', 'subject': 'ELECTRONICS', 'score': 62, 'category': 'FAILED'}
{'student': 'ALCORIZA_S5', 'subject': 'ENGLISH', 'score': 99, 'category': 'EXCELLENT'}
{'student': 'ALCORIZA_S5', 'subject': 'MATHEMATICS', 'score': 81, 'category': 'GOOD'}
{'student': 'ALCORIZA_S5', 'subject': 'PHYSICS', 'score': 64, 'category': 'FAILED'}
{'student': 'ALCORIZA_S5', 'subject': 'PROGRAMMING', 'score': 84, 'category': 'GOOD'}
PERFORMANCE SUMMARY
Total records: 27
Valid scores: 26
Incomplete records: 1


PS C:\Users\rowena\Documents\ALCORIZA\Lab4_DataStructures_Alcoriza\Lab4_Addtional_Exercises> python exercise2_studentperformace.py
{'student': 'ALCORIZA_S2', 'subject': 'PROGRAMMING', 'score': 66, 'category': 'FAILED'}
{'student': 'ALCORIZA_S3', 'subject': 'ELECTRONICS', 'score': 80, 'category': 'GOOD'}
{'student': 'ALCORIZA_S3', 'subject': 'ENGLISH', 'score': 99, 'category': 'EXCELLENT'}
{'student': 'ALCORIZA_S3', 'subject': 'MATHEMATICS', 'score': 88, 'category': 'GOOD'}
{'student': 'ALCORIZA_S3', 'subject': 'PHYSICS', 'score': 77, 'category': 'PASSED'}
{'student': 'ALCORIZA_S3', 'subject': 'PROGRAMMING', 'score': 70, 'category': 'FAILED'}
{'student': 'ALCORIZA_S4', 'subject': 'ELECTRONICS', 'score': 79, 'category': 'PASSED'}
{'student': 'ALCORIZA_S4', 'subject': 'ENGLISH', 'score': 62, 'category': 'FAILED'}
{'student': 'ALCORIZA_S4', 'subject': 'MATHEMATICS', 'score': 82, 'category': 'GOOD'}
{'student': 'ALCORIZA_S4', 'subject': 'PHYSICS', 'score': 60, 'category': 'FAILED'}
{'student': 'ALCORIZA_S4', 'subject': 'PROGRAMMING', 'score': 90, 'category': 'EXCELLENT'}
{'student': 'ALCORIZA_S5', 'subject': 'ELECTRONICS', 'score': 62, 'category': 'FAILED'}
{'student': 'ALCORIZA_S5', 'subject': 'ENGLISH', 'score': 99, 'category': 'EXCELLENT'}
{'student': 'ALCORIZA_S5', 'subject': 'MATHEMATICS', 'score': 81, 'category': 'GOOD'}
{'student': 'ALCORIZA_S5', 'subject': 'PHYSICS', 'score': 64, 'category': 'FAILED'}
{'student': 'ALCORIZA_S5', 'subject': 'PROGRAMMING', 'score': 84, 'category': 'GOOD'}
PERFORMANCE SUMMARY
Total records: 27
Valid scores: 26
Incomplete records: 1
Repeated student-subject pairs: 2
Students: 5
Subjects: 5
PS C:\Users\rowena\Documents\ALCORIZA\Lab4_DataStructures_Alcoriza\Lab4_Addtional_Exercises>

EXERCISE 3



```
# exercise 3 - transaction history analyzer

LAST_NAME = "ALCORIZA"
STUDENT_ID = "TUPM-26-1085"
FAVORITE_ARTIST = "OLIVIA RODRIGO"
SEED_NUM = 5

from collections import Counter, defaultdict


customers = [f"{LAST_NAME}_C{i}" for i in range(1, 6)]

prices = {
    "NOTEBOOK": 35.00,
    "PEN": 10.00,
    "USB": 250.00,
    "HEADPHONES": 350.00,
    "CALCULATOR": 180.00
}

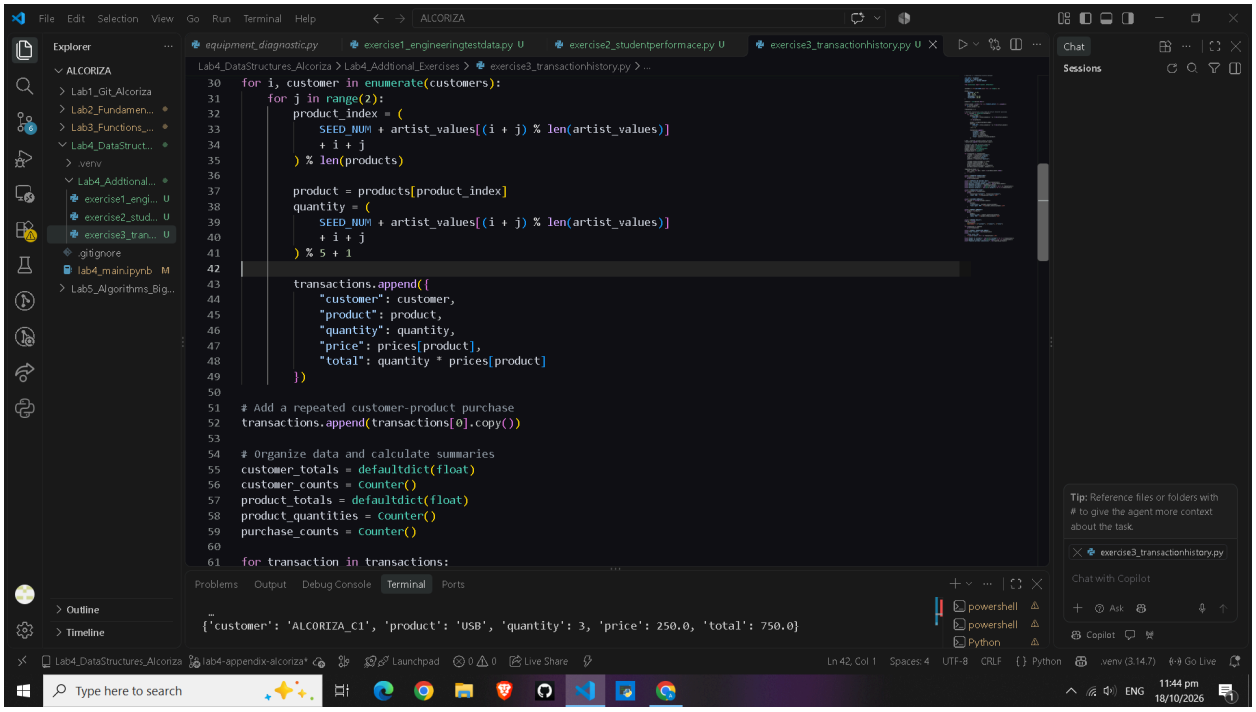
products = list(prices.keys())

artist_values = [ord(c) for c in FAVORITE_ARTIST if c.isalpha()]
if not artist_values:
    artist_values = [1]

transactions = []

# Generate transactions using seed and artist character positions
for i, customer in enumerate(customers):
    for j in range(2):
        product_index = (
```

```
{'customer': 'ALCORIZA_C1', 'product': 'USB', 'quantity': 3, 'price': 250.0, 'total': 750.0}
```



```
for i, customer in enumerate(customers):
    for j in range(2):
        product_index = (
            SEED_NUM + artist_values[(i + j) % len(artist_values)]
            + i + j
        ) % len(products)

        product = products[product_index]
        quantity = (
            SEED_NUM + artist_values[(i + j) % len(artist_values)]
            + i + j
        ) % 5 + 1

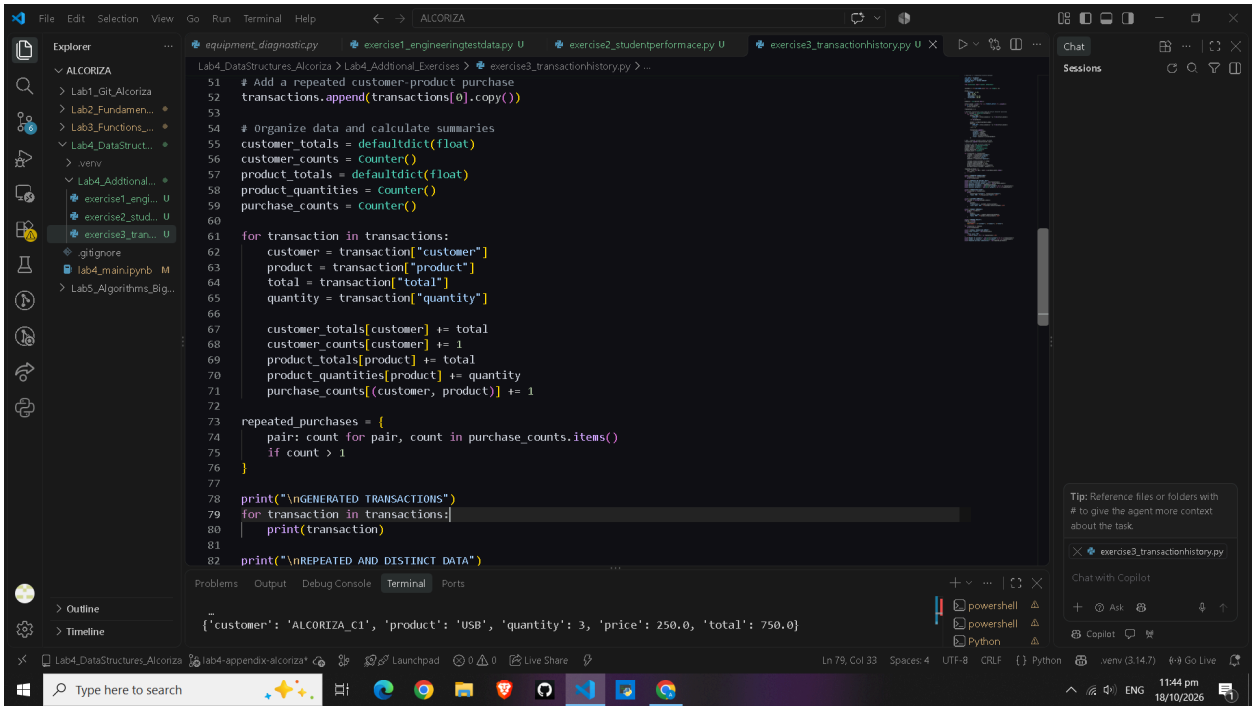
        transactions.append({
            "customer": customer,
            "product": product,
            "quantity": quantity,
            "price": prices[product],
            "total": quantity * prices[product]
        })

# Add a repeated customer-product purchase
transactions.append(transactions[0].copy())

# Organize data and calculate summaries
customer_totals = defaultdict(float)
customer_counts = Counter()
product_totals = defaultdict(float)
product_quantities = Counter()
purchase_counts = Counter()

for transaction in transactions:
```

```
{'customer': 'ALCORIZA_C1', 'product': 'USB', 'quantity': 3, 'price': 250.0, 'total': 750.0}
```



```
# Add a repeated customer-product purchase
transactions.append(transactions[0].copy())

# Organize data and calculate summaries
customer_totals = defaultdict(float)
customer_counts = Counter()
product_totals = defaultdict(float)
product_quantities = Counter()
purchase_counts = Counter()

for transaction in transactions:
    customer = transaction["customer"]
    product = transaction["product"]
    total = transaction["total"]
    quantity = transaction["quantity"]

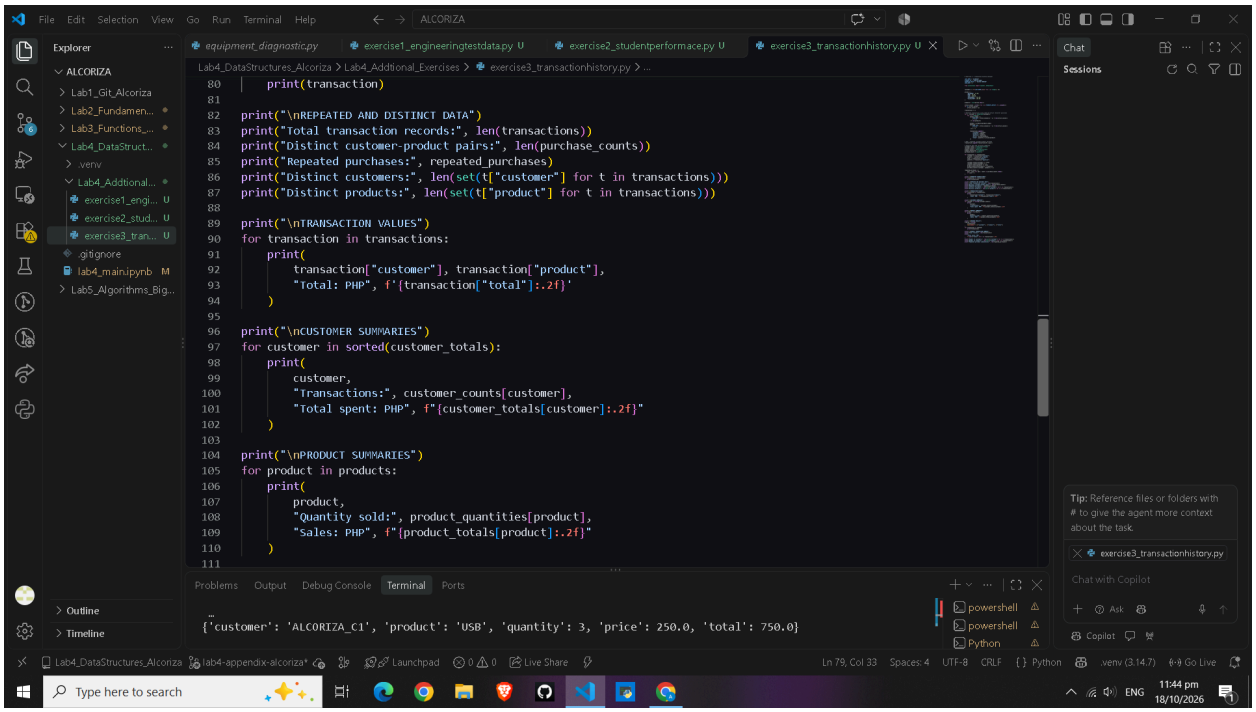
    customer_totals[customer] += total
    customer_counts[customer] += 1
    product_totals[product] += total
    product_quantities[product] += quantity
    purchase_counts[(customer, product)] += 1

repeated_purchases = {
    pair: count for pair, count in purchase_counts.items()
    if count > 1
}

print("\nGENERATED TRANSACTIONS")
for transaction in transactions:
    print(transaction)

print("\nREPEATED AND DISTINCT DATA")
```

```
{'customer': 'ALCORIZA_C1', 'product': 'USB', 'quantity': 3, 'price': 250.0, 'total': 750.0}
```



```
    print(transaction)

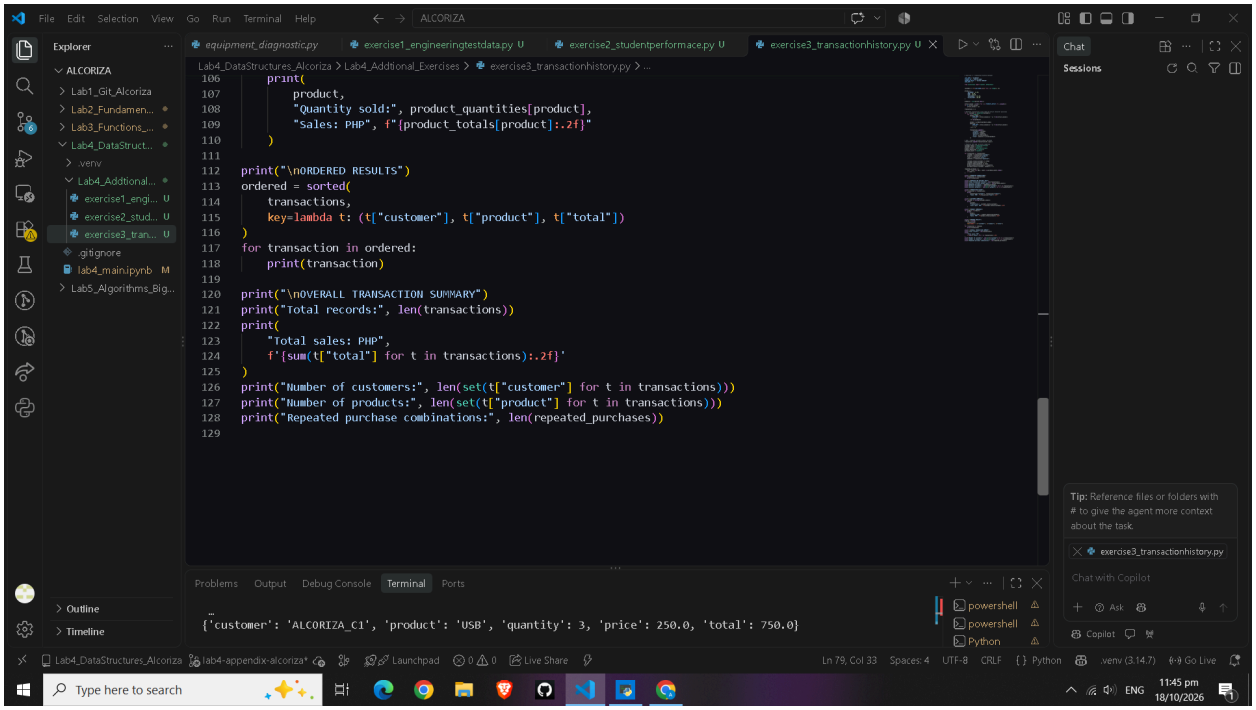
print("\nREPEATED AND DISTINCT DATA")
print("Total transaction records:", len(transactions))
print("Distinct customer-product pairs:", len(purchase_counts))
print("Repeated purchases:", repeated_purchases)
print("Distinct customers:", len(set(t["customer"] for t in transactions)))
print("Distinct products:", len(set(t["product"] for t in transactions)))

print("\nTRANSACTION VALUES")
for transaction in transactions:
    print(
        transaction["customer"], transaction["product"],
        "Total: PHP", f'{transaction["total"]:.2f}'
    )

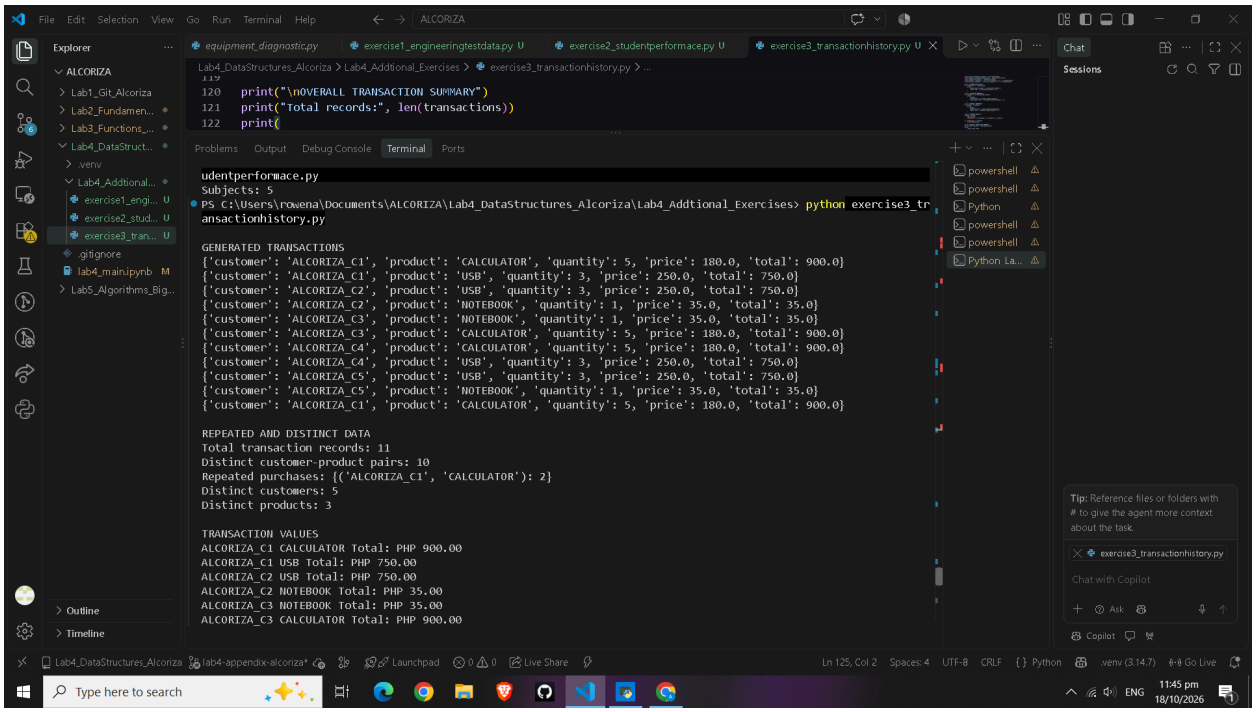
print("\nCUSTOMER SUMMARIES")
for customer in sorted(customer_totals):
    print(
        customer,
        "Transactions:", customer_counts[customer],
        "Total spent: PHP", f"{customer_totals[customer]:.2f}"
    )

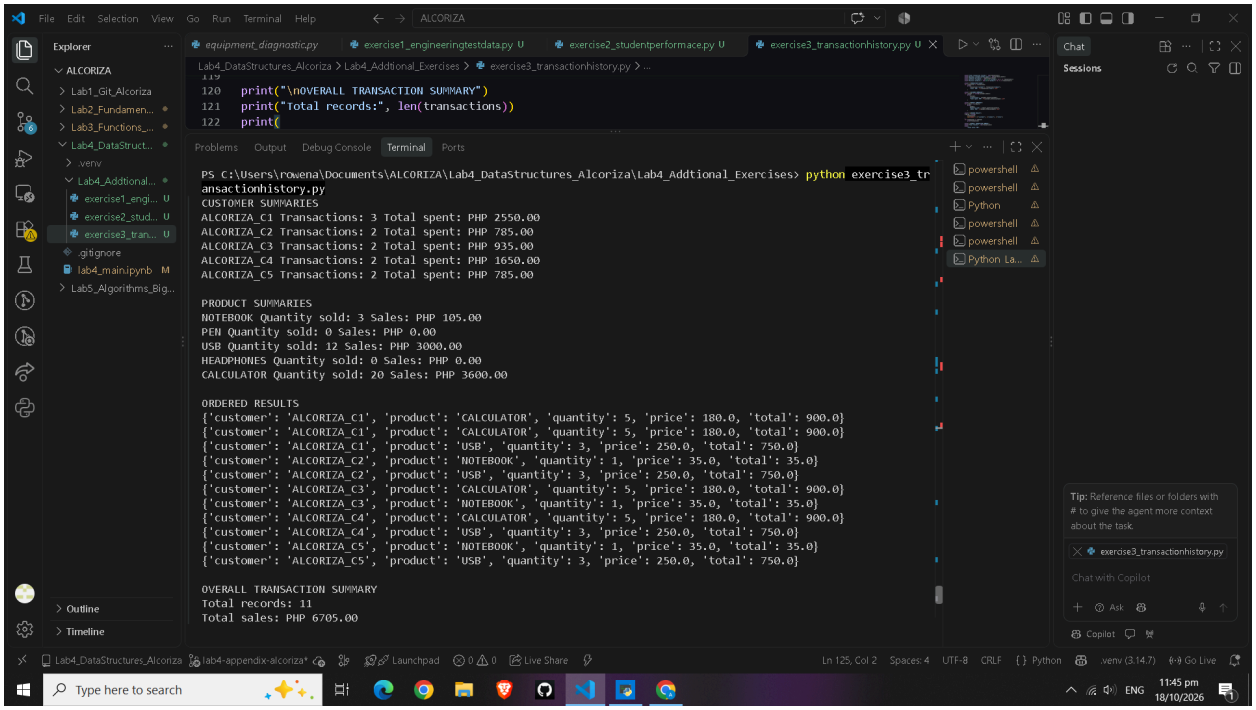
print("\nPRODUCT SUMMARIES")
for product in products:
    print(
        product,
        "Quantity sold:", product_quantities[product],
        "Sales: PHP", f"{product_totals[product]:.2f}"
    )
```

```
{'customer': 'ALCORIZA_C1', 'product': 'USB', 'quantity': 3, 'price': 250.0, 'total': 750.0}
```


print(
product,
"Quantity sold:", product_quantities[product],
"Sales: PHP", f"{product_totals[product]:.2f}"
)
print("\nORDERED RESULTS")
ordered = sorted(
transactions,
key=lambda t: (t["customer"], t["product"], t["total"])
)
for transaction in ordered:
print(transaction)
print("\nOVERALL TRANSACTION SUMMARY")
print("Total records:", len(transactions))
print(
"Total sales: PHP",
f'{sum(t["total"] for t in transactions):.2f}'
)
print("Number of customers:", len(set(t["customer"] for t in transactions)))
print("Number of products:", len(set(t["product"] for t in transactions)))
print("Repeated purchase combinations:", len(repeated_purchases))
{'customer': 'ALCORIZA_C1', 'product': 'USB', 'quantity': 3, 'price': 250.0, 'total': 750.0}

RESULTS


print("\nOVERALL TRANSACTION SUMMARY")
print("Total records:", len(transactions))
udentperformace.py
Subjects: 5
PS C:\Users\rowena\Documents\ALCORIZA\Lab4_DataStructures_Alcoriza\Lab4_Addtional_Exercises> python exercise3_tr
ansactionhistory.py
GENERATED TRANSACTIONS
{'customer': 'ALCORIZA_C1', 'product': 'CALCULATOR', 'quantity': 5, 'price': 180.0, 'total': 900.0}
{'customer': 'ALCORIZA_C1', 'product': 'USB', 'quantity': 3, 'price': 250.0, 'total': 750.0}
{'customer': 'ALCORIZA_C2', 'product': 'USB', 'quantity': 3, 'price': 250.0, 'total': 750.0}
{'customer': 'ALCORIZA_C2', 'product': 'NOTEBOOK', 'quantity': 1, 'price': 35.0, 'total': 35.0}
{'customer': 'ALCORIZA_C3', 'product': 'NOTEBOOK', 'quantity': 1, 'price': 35.0, 'total': 35.0}
{'customer': 'ALCORIZA_C3', 'product': 'CALCULATOR', 'quantity': 5, 'price': 180.0, 'total': 900.0}
{'customer': 'ALCORIZA_C4', 'product': 'CALCULATOR', 'quantity': 5, 'price': 180.0, 'total': 900.0}
{'customer': 'ALCORIZA_C4', 'product': 'USB', 'quantity': 3, 'price': 250.0, 'total': 750.0}
{'customer': 'ALCORIZA_C5', 'product': 'USB', 'quantity': 3, 'price': 250.0, 'total': 750.0}
{'customer': 'ALCORIZA_C5', 'product': 'NOTEBOOK', 'quantity': 1, 'price': 35.0, 'total': 35.0}
{'customer': 'ALCORIZA_C1', 'product': 'CALCULATOR', 'quantity': 5, 'price': 180.0, 'total': 900.0}
REPEATED AND DISTINCT DATA
Total transaction records: 11
Distinct customer-product pairs: 10
Repeated purchases: {('ALCORIZA_C1', 'CALCULATOR'): 2}
Distinct customers: 5
Distinct products: 3
TRANSACTION VALUES
ALCORIZA_C1 CALCULATOR Total: PHP 900.00
ALCORIZA_C1 USB Total: PHP 750.00
ALCORIZA_C2 USB Total: PHP 750.00
ALCORIZA_C2 NOTEBOOK Total: PHP 35.00
ALCORIZA_C3 NOTEBOOK Total: PHP 35.00
ALCORIZA_C3 CALCULATOR Total: PHP 900.00


print("\nOVERALL TRANSACTION SUMMARY")
print("Total records:", len(transactions))
print(
PS C:\Users\rowena\Documents\ALCORIZA\Lab4_DataStructures_Alcoriza\Lab4_Addtional_Exercises> python exercise3_transactionhistory.py
CUSTOMER SUMMARIES
ALCORIZA_C1 Transactions: 3 Total spent: PHP 2550.00
ALCORIZA_C2 Transactions: 2 Total spent: PHP 785.00
ALCORIZA_C3 Transactions: 2 Total spent: PHP 935.00
ALCORIZA_C4 Transactions: 2 Total spent: PHP 1650.00
ALCORIZA_C5 Transactions: 2 Total spent: PHP 785.00
PRODUCT SUMMARIES
NOTEBOOK Quantity sold: 3 Sales: PHP 105.00
PEN Quantity sold: 0 Sales: PHP 0.00
USB Quantity sold: 12 Sales: PHP 3000.00
HEADPHONES Quantity sold: 0 Sales: PHP 0.00
CALCULATOR Quantity sold: 20 Sales: PHP 3600.00
ORDERED RESULTS
{'customer': 'ALCORIZA_C1', 'product': 'CALCULATOR', 'quantity': 5, 'price': 180.0, 'total': 900.0}
{'customer': 'ALCORIZA_C1', 'product': 'CALCULATOR', 'quantity': 5, 'price': 180.0, 'total': 900.0}
{'customer': 'ALCORIZA_C1', 'product': 'USB', 'quantity': 3, 'price': 250.0, 'total': 750.0}
{'customer': 'ALCORIZA_C2', 'product': 'NOTEBOOK', 'quantity': 1, 'price': 35.0, 'total': 35.0}
{'customer': 'ALCORIZA_C2', 'product': 'USB', 'quantity': 3, 'price': 250.0, 'total': 750.0}
{'customer': 'ALCORIZA_C3', 'product': 'CALCULATOR', 'quantity': 5, 'price': 180.0, 'total': 900.0}
{'customer': 'ALCORIZA_C3', 'product': 'NOTEBOOK', 'quantity': 1, 'price': 35.0, 'total': 35.0}
{'customer': 'ALCORIZA_C4', 'product': 'CALCULATOR', 'quantity': 5, 'price': 180.0, 'total': 900.0}
{'customer': 'ALCORIZA_C4', 'product': 'USB', 'quantity': 3, 'price': 250.0, 'total': 750.0}
{'customer': 'ALCORIZA_C5', 'product': 'NOTEBOOK', 'quantity': 1, 'price': 35.0, 'total': 35.0}
{'customer': 'ALCORIZA_C5', 'product': 'USB', 'quantity': 3, 'price': 250.0, 'total': 750.0}
OVERALL TRANSACTION SUMMARY
Total records: 11
Total sales: PHP 6705.00